

IMPERIAL

Sonar-Based Underwater SLAM with 3D Gaussian Splatting Mapping

**Online-render map-pose registration
for Pose3 correction**

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Final Year Project Presentation

Why Underwater Sonar SLAM Is Hard

Can a renderable sonar map provide online correction factors?

1. Problem setting

Underwater robots cannot use GNSS. IMU/DVL odometry therefore drifts over time.

2. Sensor limitation

Sonar works underwater, but the image evidence is sparse, ambiguous, and viewpoint-dependent.

3. Classical SLAM limitation

Geometric matching becomes fragile under sonar noise and repeated acoustic structure.

4. Key insight from 3DGS

A learned Gaussian scene can predict the sonar view from a candidate pose.

5. Gap

3DGS is rarely used as an online sonar measurement model inside SLAM.

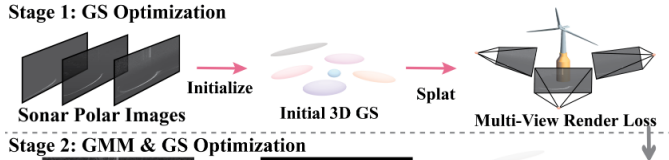
6. Project idea

Use render consistency between live sonar and NAS-GS prediction as a Pose3 SLAM constraint.

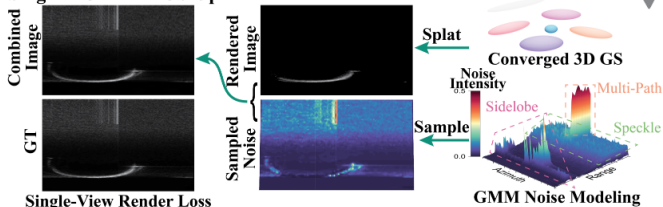
NAS-GS: Noise-Aware Sonar Gaussian Splatting

A renderable soanr measurement model

Stage 1: GS Optimization



Stage 2: GMM & GS Optimization



NAS-GS training pipeline, adapted from Xu et al.

What NAS-GS does

- Learns a 3D Gaussian sonar scene from polar sonar images.
- Optimises rendered structure together with acoustic noise appearance.
- Produces a differentiable sonar renderer for new viewpoints.

How I use it

- Freeze the trained map during replay.
- Render candidate Pose3 hypotheses online.
- Insert a factor only when live sonar and render agree.

Render Consistency As A Pose3 Factor

NAS-GS predicts observations; iSAM2 consumes accepted constraints

Online loop: render from the map, compare with the live frame, then add a graph factor only when the evidence is strong.

Renderable sonar map

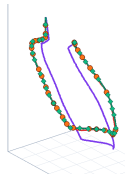
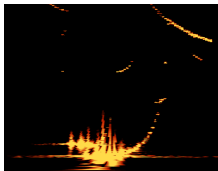
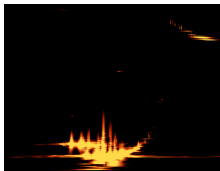
NAS-GS predicts what the sonar should observe from a proposed Pose3 camera pose.

Online evidence

A live sonar frame is compared against rendered candidates around the current drifted pose.

Graph correction

If render similarity and gates pass, the recovered pose is inserted as a Pose3 factor.

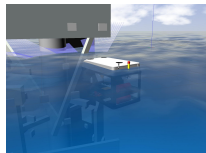
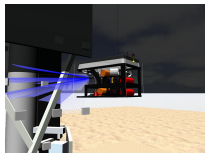


Simulation Platform: DAVE In Docker

Repeatable sonar, IMU, DVL, and ground-truth recording

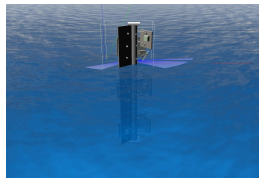
Docker setting and robot model

- CUDA-enabled Docker container.
- DAVE with ROS Noetic and Gazebo.
- BlueROV2 vehicle model.
- Sensors recorded: sonar, IMU, DVL, odometry, and ground truth.



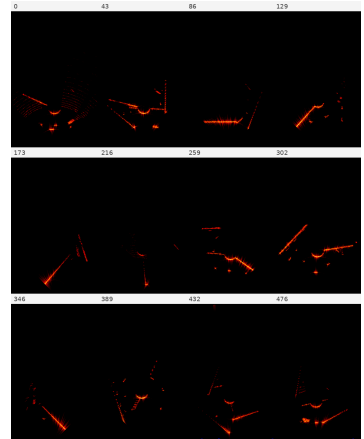
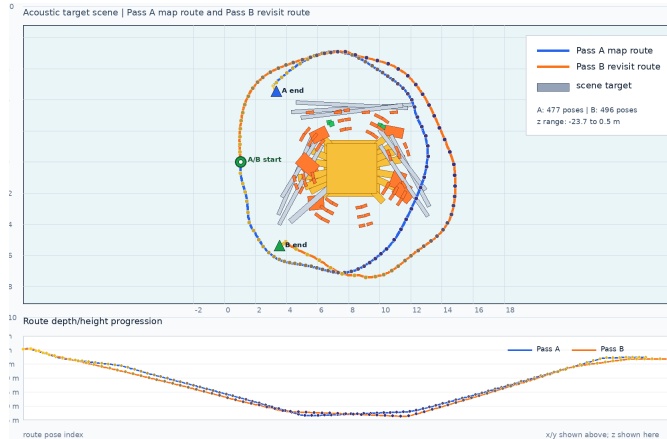
Custom scene description

- A fixed underwater target structure creates repeatable sonar returns.
- Pass A, Pass A revisit, and Pass B use controlled waypoint routes.
- Sonar field of view, range, beam pattern, and noise are explicitly set.



Pass A Map Recording And Revisit Path

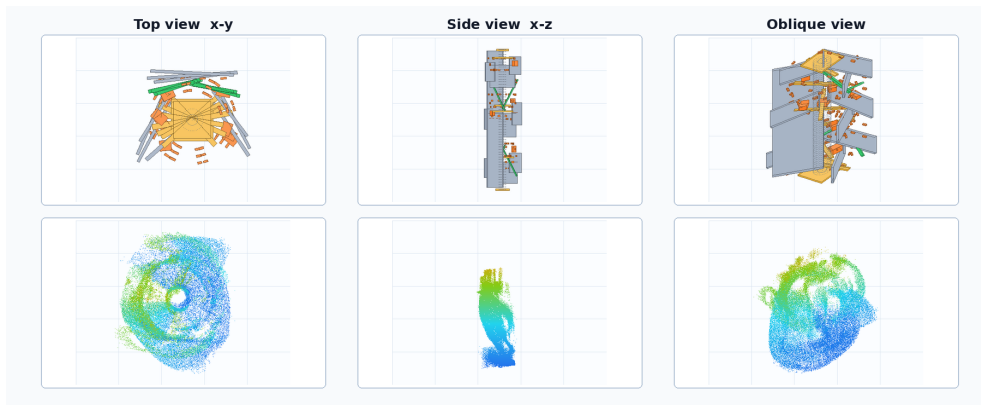
Pass A trains the NAS-GS map; Pass B reuses the start point but follows a harder revisit path



From Simulated Geometry To NAS-GS Map

The learned map is sonar-visible structure, not a full CAD reconstruction

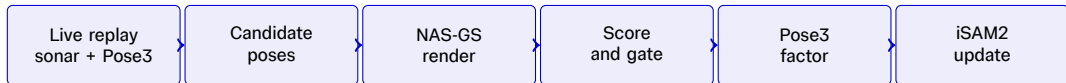
The two rows use the same top, side, and oblique projections. They differ because the DAVE row shows the full simulated target geometry, while the NAS-GS row shows Gaussian centres learned only from sonar-visible acoustic returns.



Top row: DAVE target geometry from SDF. Bottom row: NAS-GS map centres from point_cloud.ply. Same projection and bounds per column.

Online-render Registration Backbone

Same online loop for same-route correction and revisit correction



Local single

Compact x, y, z, ψ search around the live Pose3 baseline. Main same-route policy for Pass A and off-shore replay.

Route-progress revisit

Sequence render search guided by route progress and drift-prior checks. Used for revisit cases such as Pass B.

Gating is separate from the mode. Relaxed gating accepts useful factors earlier, while score and structure checks still prevent weak render matches from entering the graph.

Algorithm

Candidate poses are searched first; the graph is changed only after gating

Candidate poses

- Start from the live IMU/DVL Pose3 estimate.
- Sample small corrections in x , y , z and yaw.
- Local mode searches a compact recoverable basin.
- Revisit mode uses a wider sequence proposal guided by route progress and drift prior.
- A candidate is always a possible robot pose. The NAS-GS map is only used to render evidence.

Score and gate

- Each candidate is rendered by the fixed NAS-GS map.
- Image score combines NCC-style similarity with error after intensity normalisation.
- Structure score checks that bright sonar returns and rendered returns overlap.
- Gates reject low score, weak structure, excessive pose jump, and unstable sequence support.
- The best passing candidate provides the Pose3 factor pose and covariance.

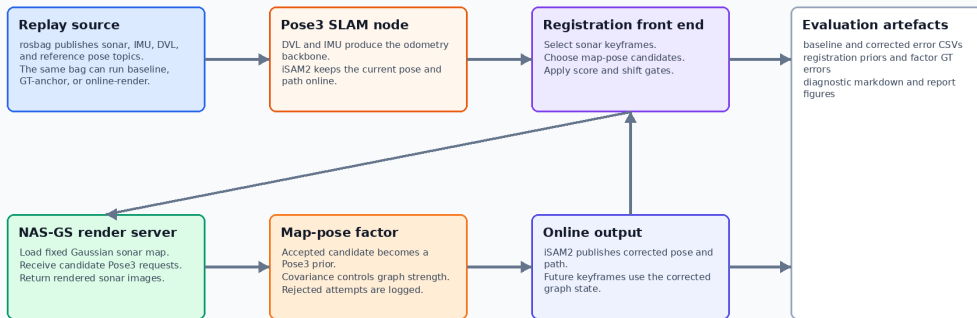
The key design choice is conservative evidence: a render match can correct drift, but a failed match does not damage the pose graph.

Live Replay Implementation

The same replay path is used for synthetic and offshore validation

ROS live replay pipeline for online-render SLAM

The replay is live in the estimator. Accepted render factors immediately affect the following iSAM2 trajectory.



This is why live replay differs from offline plotting. Once a factor is accepted, the backend state changes before later sonar frames arrive. The replay therefore approximates online SLAM behaviour while still keeping a readable bag and evaluation files.

Experimental Design

Three increasingly difficult operating cases

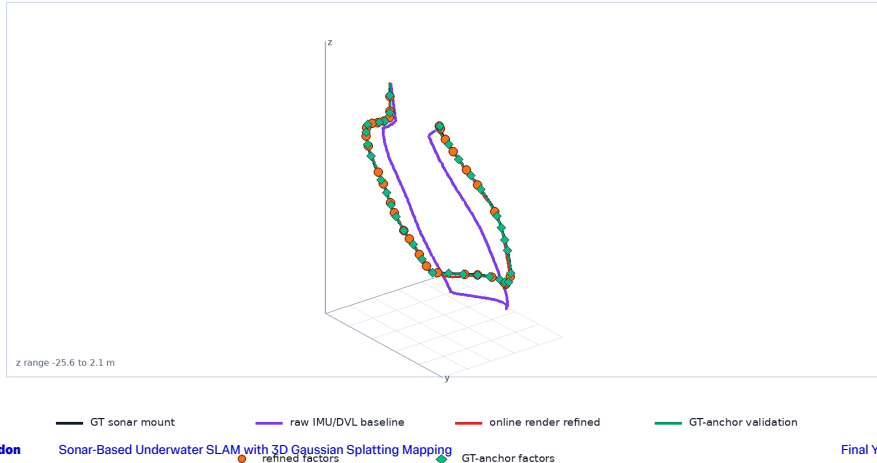
Case	Purpose	Sonar setting
Pass A	Same-route correction on the trained DAVE map	90° field of view, 20 m range
Pass A revisit	Same-route revisit with a newly recorded trajectory	90°, 20 m
Pass B revisit	Hard revisit around the scene in a different direction	90°, 20 m
Offshore	Real-scene style validation on the 151216 dataset	120°, 12 m

The baseline is the IMU/DVL Pose3 replay. The corrected result is the same backend after NAS-GS registration factors are inserted.

Pass A Result

The same-route case shows the strongest online-render correction

Pass A 3D z-drift diagnostic with map-pose registration



Pass A Policy Selection

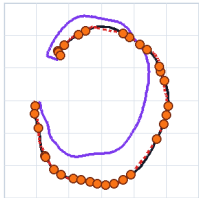
Rendered BP library with runtime no-BP was the most reliable local policy

Pass A online-render with BP and no-BP rendering

Same relaxed local single policy with different NAS-GS render settings

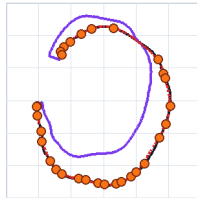
BP90x20 rendering

36 factors XY RMSE 2.741 to 0.733 m mean |z| 12.719 to 0.901 m



no-BP rendering

31 factors XY RMSE 2.741 to 0.612 m mean |z| 12.719 to 0.467 m



— GT sonar mount

— baseline

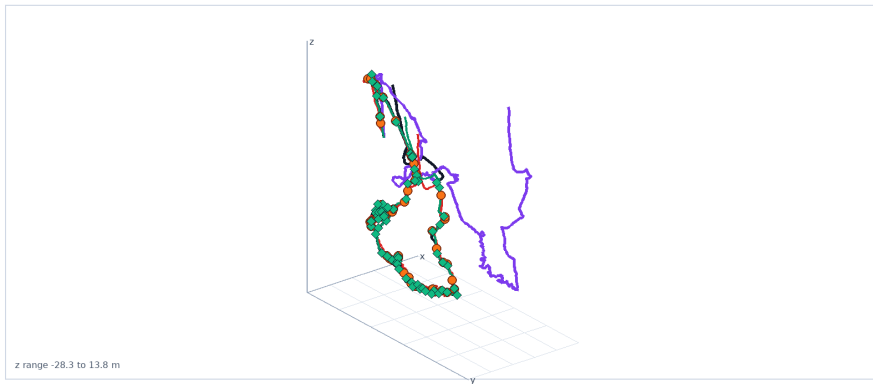
- - - corrected

● accepted NAS-GS factor

Offshore Validation

The method transfers, but the real-scene case exposes z and appearance limits

Offshore 3D pose graph with live online-render correction



— GT sonar mount

— raw IMU/DVL baseline

— online render refined

— GT-anchor validation

● refined factors

◆ GT-anchor factors

BP versus no-BP on Offshore Data

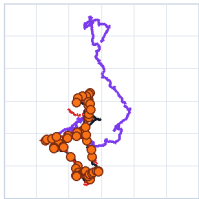
The better rendering mode depends on the recorded sonar appearance

Offshore online-render with BP and no-BP rendering

Same relaxed local single policy on real sonar with different NAS-GS render settings

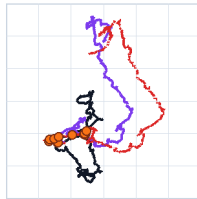
BP rendering

65 factors XY RMSE 11.398 to 1.107 m mean |z| 6.129 to 0.693 m



no-BP rendering

12 factors XY RMSE 11.398 to 12.550 m mean |z| 6.129 to 1.261 m



— GT sonar mount

— baseline

— corrected

● accepted NAS-GS factor

Pass A is cleaner and benefits from the rendered BP library even when runtime BP is disabled. Offshore data contains stronger real-image artefacts, so BP/no-BP behaviour is less interchangeable.

Revisit Baselines

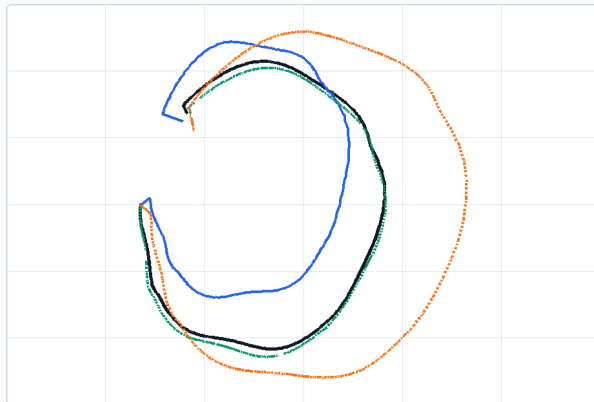
Pass A revisit is same-route; Pass B is harder because the route direction changes

Pass A and Pass A revisit realistic-drift baselines

Top-view uses the graph baseline. The z panel uses raw IMU/DVL odometry, where baseline spans 24.7 m and 24.8 m against GT spans 23.9 m and 23.9 m.

Pass A revisit is yaw-aligned to original Pass A for display only. Alignment yaw 70.1 deg.

top-view trajectory in online map frame



z (m)

2.9

z profile over normalised route progress

-4.4

-11.6

-18.9

-26.10

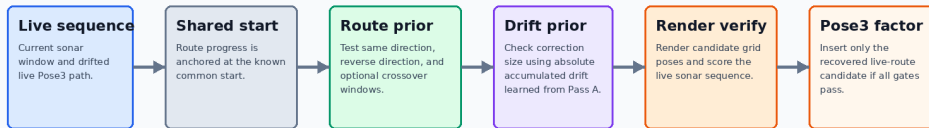


Route-progress Revisit Mode

Sequence evidence is used before a revisit factor is accepted

Route-progress online-render revisit policy

Live sonar sequence evidence is used before a Pass B or revisit factor is inserted.



Mathematical role

Candidate poses are sampled around the current drifted live pose. For each candidate c the score is $J(c) = \text{mean } s(l, R(c))$ minus drift, route-progress, edge, and smoothness penalties. The Pass A map is only the renderable reference. The accepted factor pose is the recovered live-route candidate.

Revisit Result

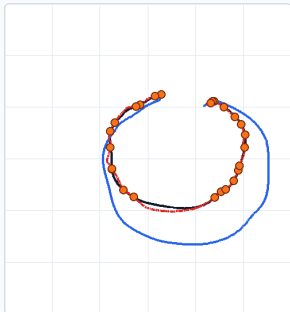
The revisit policy improves the hard case conservatively

Route-progress and same-route revisit replay results

Local single is strong for the same-route Pass A revisit. Route-progress gives guarded partial correction for the harder revisit cases.

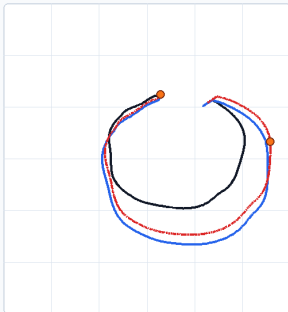
Pass A revisit local single

26 factors XY 2.901 to 1.008 m mean $|z|$ 12.123 to 1.034 m



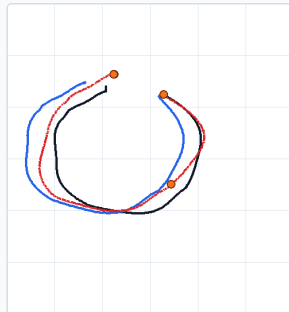
Pass A revisit route-progress

2 factors XY 2.901 to 2.442 m mean $|z|$ 12.123 to 1.946 m



Pass B route-progress

4 factors XY 2.734 to 1.875 m mean $|z|$ 13.115 to 3.605 m



— GT sonar mount

— baseline

- - - corrected

● accepted NAS-GS factor

What The Results Show

Strengths

- Local same-route correction works.
- The pipeline runs as a live replay.
- Offshore validation shows real-scene relevance.
- Render evidence gives interpretable factor placement.

Limits

- Sonar ambiguity can accept visually plausible but wrong candidates.
- z correction is sensitive to rendering and trajectory priors.
- Hard revisit remains limited without stronger sequence geometry.
- Render-server geometry must match the sonar dataset.

Conclusion

This project demonstrates an online-render SLAM correction system.

The main result is that a NAS-GS sonar map can provide useful Pose3 registration factors when the live sonar appearance, render geometry, and search basin are aligned.

The main limitation is revisit robustness. The next step is a stronger sequence-level solver that estimates smooth candidate paths rather than accepting isolated frame evidence.

